

Seohyeon Chang

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Research Interests

Robot Learning and Control
Geometric Deep Learning
Optimization on Manifolds

Research Experience

Dynamic Robot Control and Design Lab — Jul. 2026 – Present
Undergraduate Research Intern, KAIST — Daejeon, South Korea
Advisor: Prof. Hae-Won Park

Education

Korea University — Mar. 2025 – Present
B.S. Candidate in Computer Science and Engineering; Double Major in Mechanical Engineering — GPA: 4.45/4.50
Seoul, South Korea

Chungbuk Science High School — Mar. 2023 – Feb. 2025
Graduated early
Chungbuk, South Korea

Selected Projects

Selective Progressive Adapter for Refusal Calibration in Language Models — May 2026
Course Research Project, COSE461 — LLM Safety

- Analyzed refusal representations in Llama-3.2-3B-Instruct to study jailbreak vulnerability and over-refusal.
- Built a progressive mixture-of-experts adapter for selective representation-level safety correction.

Distributed Quantum Machine Learning — May 2026 – Present
Research Project, AIKU — Quantum Machine Learning

- Studied how pre-shared Bell-pair entanglement can recover cross-party information in distributed QML.
- Designed a residual Fourier-correlation pipeline to select useful entanglement pairs across QPUs.

Reinforcement Learning for Robot Object Pushing — Feb. 2026
Project, AIKU — Robot Learning

- Built a custom Webots–Gymnasium environment for training an e-puck robot to push an object toward a target.
- Trained PPO policies with reward shaping based on robot–object distance, target progress, and pushing alignment.

AI Storyboard Sketch Generator — Dec. 2025
Project, AIKU — Generative AI

- Developed an AI pipeline for generating storyboard sketches from scene descriptions & camera-shot conditions.
- Fine-tuned Stable Diffusion with textual embeddings and data augmentation to improve composition control.

Honors and Awards

Dean's List, High Honors Spring 2025, Fall 2025
Korea University; semester GPA $\geq$ 4.00/4.50

Dean's Award for Academic Excellence Spring 2025
Korea University; perfect semester GPA of 4.50/4.50

ICPC 2025 Seoul Regional Nov. 2025
49th place; solved 6 problems

ICPC 2025 Seoul Online Preliminary Contest Oct. 2025
51st place; solved 4 problems

Leadership and Academic Activities

AIKU, Deep Learning Club Jul. 2025 – Present
Vice President, Korea University

- Organize academic sessions, technical seminars, and collaborative projects for the university deep learning community.
- Delivered selected talks on ML/DL basics, PyTorch, group-equivariant CNNs, language-model arithmetic, image tokenization, and SLAM.
- Organized the Yonsei–Korea Joint AI Hackathon and the **11th AIKUTHON** Kaggle competition.
- Served as a 2025 Google Student Ambassador, #TeamGemini.

ALPS, Algorithm & Problem Solving Club Sep. 2025 – Jan. 2026
President, Korea University

- Led regular algorithm study sessions and managed club operations as president.
- Organized the **2025 Korea University Collegiate Programming Contest (KCPC)**, managing logistics and serving as a problem setter and reviewer.

MatKor, Information Security & Algorithm Club May 2025 – Jan. 2026
Executive Board Member, Korea University

- Served as a problem setter and reviewer for **The 7th MatKor Cup**.

Skills

Programming: C, C++, Python, NumPy, PyTorch
Technical Areas: Machine learning, algorithmic problem solving
Tools: LaTeX, Markdown, Git
Languages: Korean native, English, Chinese HSK 4